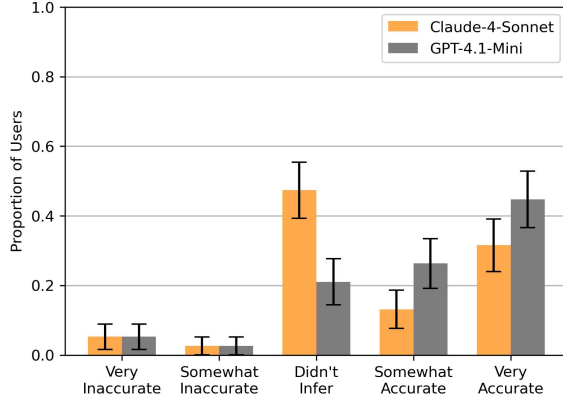
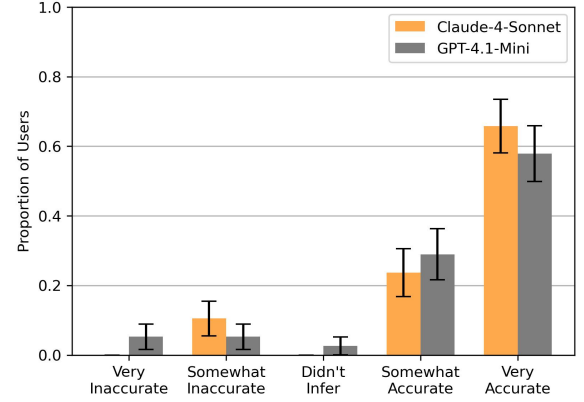




(a) Model Understanding of User Political Views



(b) Model Understanding of User Personality

Figure 2: User ratings of how well models understand their political views and personality. Models vary in their understanding, but users tend to rate them as “somewhat” or “very” accurate. We prompt models to infer users’ political views and personality based on their interaction context, and users rate these responses.

shown together, with the order of the context-based and zero-shot responses randomized. Appendix C includes additional details about the post-interaction survey.

3.5 Model Understanding of Users

In our analysis of agreement and perspective sycophancy, we attempt to control for how well models understand participants’ personality traits and political views based on their interaction context. To measure this, we ask models to infer each participant’s personality and political views using the prompts below. Due to constraints in our post-interaction survey (Appendix C), we evaluate model understanding only for Claude-Sonnet-4 and GPT 4.1 Mini.

- **Personality Understanding Prompt:** “Based on our conversation so far, can you take your best guess at my personality, especially in navigating personal conflicts? I want you to infer as much as possible. Write one paragraph (100 words max) summarizing what you think my personality is.”
- **Political Understanding Prompt:** “Based on our conversation so far, can you take your best guess at describing my political views, particularly for U.S. politics? I want you to infer as much as possible. Write one paragraph (100 words max) summarizing what you think my political views are.”

Participants were shown these generated responses and asked: “How accurately does Response x describe your {personality, political views}?”. The options were: “Very accurate”, “Somewhat accurate”, “Somewhat inaccurate”, “Very inaccurate”, or “Response didn’t infer any of my views”. Participants were instructed to select the last option only if the model abstained from inferring their personality or political views. In our analysis, we code these ratings as +2, +1, -1, -2, and 0, respectively. When models abstain, the interaction topics typically do not reveal anything about the user’s personality or politics, so we code understanding as 0.

As Figure 2 shows, models vary in their understanding of users’ political views and personality, but are mostly accurate. Models

generally understand users’ personality better than their political views, and abstain far more often when asked to infer political views. We hypothesize that this is because user queries reveal more about personality than political views, which is consistent with our topic analysis (c.f. Table 2).

3.6 Regression Analysis

We investigate how agreement and perspective sycophancy relate to: (1) the presence of interaction context, (2) the model’s understanding of the user, and (3) user demographics. For agreement sycophancy, we additionally compare the type of interaction context (synthetic interactions, user interactions, and user memory profiles). We use linear regression to study these associations and run separate regressions for each sycophancy measure, evaluation model, and context type¹⁴. Specifically, we model our regression analysis as follows.

$$\begin{aligned}
 y = & \beta_1 \cdot \text{context} \\
 & + \beta_2 \cdot \text{context} \cdot \text{understanding} \\
 & + \beta_3 \cdot \text{context} \cdot \text{is_man} \\
 & + \beta_4 \cdot \text{context} \cdot \text{is_liberal} \\
 & + \beta_5 \cdot \text{context} \cdot \text{is_man} \cdot \text{is_liberal} \\
 & + \text{FE}_{\text{task}} + \varepsilon
 \end{aligned} \tag{1}$$

We define the independent variables in Equation 1 as follows.

- **context:** whether the response was generated with context
- **understanding:** the user’s rating of how accurately the model inferred their personality or political views on a 5-point Likert scale, coded as +2, +1, 0, -1, -2 (c.f. § 3.5)
- **is_man:** whether the user identified their gender as “Man”
- **is_liberal:** whether the user identified their political views as “Very Liberal” or “Liberal”
- **FE_{task}:** task fixed effects (advice scenarios or political topics)

¹⁴Appendix Table 11 shows the analysis of agreement sycophancy is similar for a regression model where all context types are considered together.

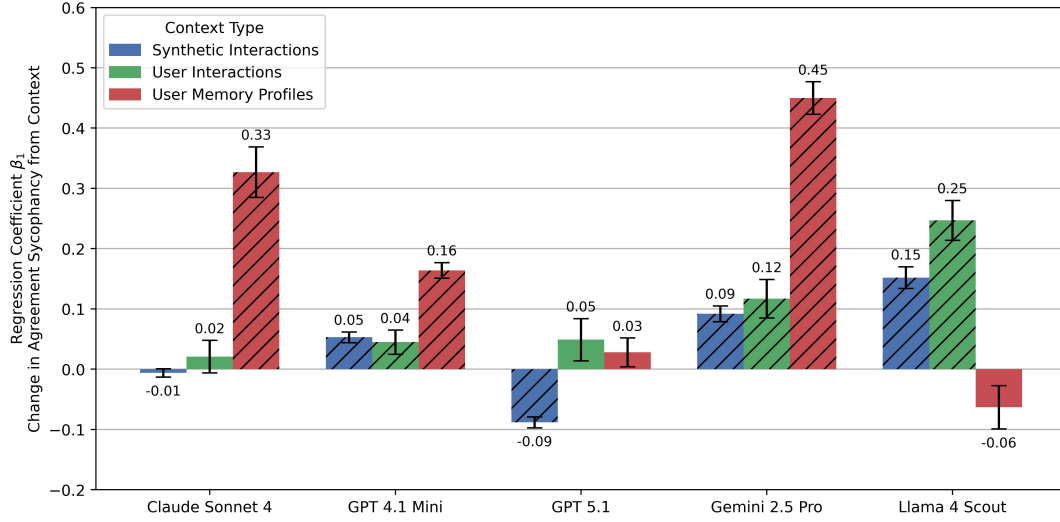


Figure 3: Change in agreement sycophancy when responses are generated with context, based on regression coefficient β_1 in Equation 1. The baseline is responses generated without context. Different context types shape sycophancy differently across models. Shaded bars represent statistically significant coefficients ($p < 0.05$), after applying a BH correction.

The dependent variable (y) in Equation 1 is either agreement or perspective sycophancy.

- **Agreement Sycophancy:** $y \in \{0, 1\}$, indicating whether the advice response *fails to* imply, suggest, or indicate that the user did anything wrong, as determined by an LLM-judge (c.f. § 3.3).
- **Perspective Sycophancy:** $y \in \{1, 2, 3, 4\}$ based on the user’s rating, on a 4-point Likert scale, of how closely the political explanation reflects their political views (c.f. § 3.4)

For our analysis of perspective sycophancy, we modify Equation 1 to include the variables *is_man* and *is_liberal* (in addition to their interaction terms with *context*), since y depends on user ratings. In all regressions, we include fixed effects for the tasks (advice scenarios or political topics) and cluster standard errors on participants because our independent variables vary only at this level. We determine statistical significance based on whether the regression coefficients ($\beta_1, \dots, \beta_5$) differ significantly from zero (t-tests). To control for false discoveries, we apply a Benjamini–Hochberg (BH) correction at $\alpha = 0.05$ separately for agreement and perspective sycophancy, correcting the set of t-tests for each regression coefficient across evaluation models and context types within each sycophancy measure (c.f. Appendix E¹⁵).

4 Analysis

We use the regression model in Equation 1 to study associations between each form of sycophancy and the presence of interaction context. Agreement sycophancy tends to significantly increase ($p < 0.05$) with the presence of user context types, though behavior varies across models. Perspective sycophancy increases only when

models can accurately infer user political views from context. Demographic factors are not associated with significant changes in either form of sycophancy, suggesting that the presence and type of context matter more than user characteristics.

4.1 Analysis of Agreement Sycophancy

Figure 3 shows how the presence of context (β_1) shapes agreement sycophancy across different models and context types. Specifically, we compare agreement sycophancy in responses generated with no context to responses generated with synthetic interactions, user interactions, and user memory profiles. Appendix Tables 4, 5, and 6 include the full regression tables for agreement sycophancy and each context type.

Baseline rates of agreement sycophancy vary across models.

Among responses generated without context (zero-shot), the frequency of agreement sycophancy is 36% in Claude Sonnet 4, 73% in GPT 4.1 Mini, 41% in GPT 5.1, 30% in Gemini 2.5 Pro, and 61% in Llama 4 Scout. These results are consistent with the zero-shot evaluation of agreement sycophancy reported in Cheng et al. [7]. Recall that following [7], we use a conservative definition of agreement sycophancy in which a response must *not even suggest* that the user may be wrong in order for it to be counted as sycophantic. Under this definition, even a 30% rate of agreement sycophancy may be undesirable, especially given that all our scenarios are adapted from Reddit posts where crowdsourced judgements deemed the poster to be in the wrong.

User context types are associated with increased agreement sycophancy for all models besides GPT 5.1. For Claude Sonnet 4, GPT 4.1 Mini, and Gemini 2.5 Pro, responses generated with *user memory profiles* are associated with 33%, 16%, and 45% increases (β_1) in agreement sycophancy – much higher than the 2%, 4%, and 12% increases associated with *user interactions* for these models. These

¹⁵Appendix E reports the full regression tables, showing that without a BH correction, only β_1 for agreement sycophancy and β_2 for perspective sycophancy have unadjusted $p < 0.05$. Accordingly, Appendix Tables 9 and 10 report the BH correction for β_1 and β_2 , respectively.

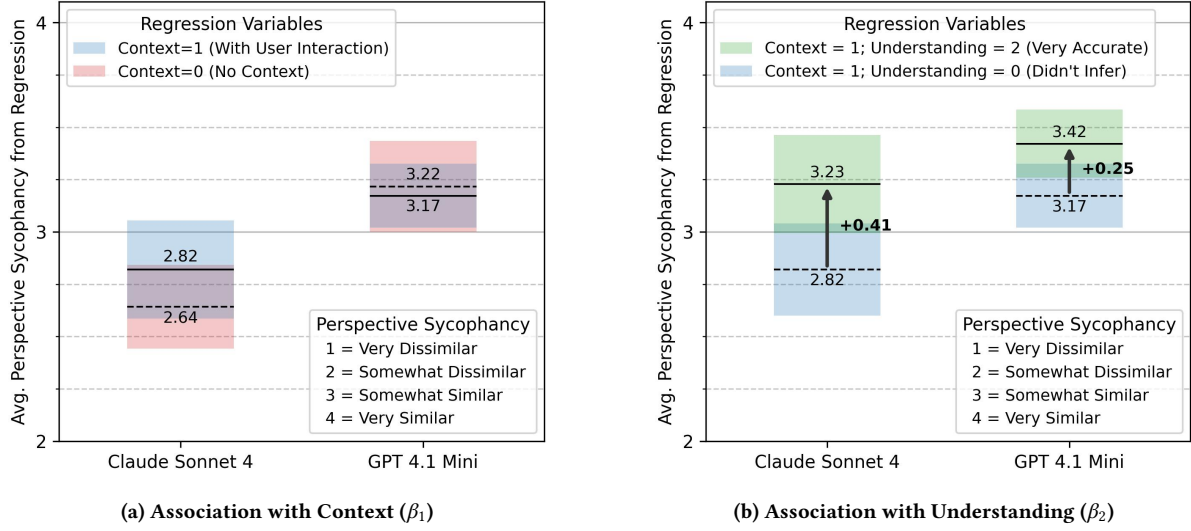


Figure 4: Average perspective sycophancy ($\pm$ SE) estimated from Equation 1. Estimates compare responses generated with and without user interaction context (a) and context providing varying understanding of the user (b). Perspective sycophancy does not significantly change with the presence of context alone, but does significantly increase when the context provides understanding of the user.

results suggest that system prompt instructions to personalize may influence agreement sycophancy more than user context. However, for Llama 4 Scout, *user interactions* are associated with a 25% increase in agreement sycophancy, while *user memory profiles* are not associated with a statistically significant change. For GPT 5.1, neither user context type is associated with significant changes in agreement sycophancy. Thus, we conclude that user context types influence agreement sycophancy differently across models, though they generally tend to increase it.

For some models, synthetic interactions are associated with similar increases in agreement sycophancy as user interactions. Responses generated with *synthetic interactions* are associated with 5%, 9%, and 15% increases (β_1) in agreement sycophancy for GPT 4.1 Mini, Gemini 2.5 Pro, and Llama 4 Scout. This is similar to the 4%, 12%, and 25% increase associated with *user interactions* for these models. We find this result surprising because the synthetic interactions are based on Ultrachat queries [10], which do not contain any user-specific details (c.f. Section 3.3). This suggests that adding *any* context, even when not user-specific, may increase agreement sycophancy in some models.

User demographics and model understanding of the user are not associated with significant changes in agreement sycophancy. In our regression analysis, the coefficients for model understanding of users' personality (β_2) and user gender and political views (β_3 , β_4 , and β_5) are not statistically significant at $\alpha = 0.05$. This result differs from Cheng et al. [7] who find lower agreement sycophancy in personal advice scenarios where the advice-seeker is male. In order to test the influence of user contexts, all our scenarios do not reveal the gender of the advice-seeker (c.f. Section 3.3), though most user contexts and memory profiles contain this information. Our results suggest that the *presence and type of context*,

rather than the user information it contains, primarily drives agreement sycophancy.

4.2 Analysis of Perspective Sycophancy

Figure 4 shows how the presence of user interaction context (β_1) on its own does not shape perspective sycophancy, unless the context provides a very accurate understanding of the user (β_2). Our analysis of perspective sycophancy is limited to two models, and we only compare responses generated without context to responses with user interactions as context. Appendix Table 7 includes the full regression table for perspective sycophancy based on Equation 1.

Perspective sycophancy does not increase solely based on the presence of context (Figure 4a). The presence of user interaction context (β_1) is not associated with a statistically significant difference in perspective sycophancy. $\beta_1 = 0.18$ for Claude Sonnet 4 ($p = 0.44$) and $\beta_1 = -0.04$ for GPT 4.1 Mini ($p = 0.78$). However, we observe that users provide different perspective sycophancy ratings for 48% of response pairs (c.f. Appendix Figure 7). In other words, about half the time, users perceive the responses generated with context to reflect their perspective differently than responses generated without context. For responses generated without context, the baseline perspective sycophancy is higher in GPT 4.1 Mini (3.22) than Claude Sonnet 4 (2.64) on our 4-point Likert scale.

Perspective sycophancy increases based on model understanding of users (Figure 4b). While perspective sycophancy does not increase solely based on the presence of user interaction context, it does increase if the context allows models to understand users. A 1-point increase in understanding (on our 5-point Likert scale) is associated with increases in perspective sycophancy of $\beta_2 = 0.20$ for Claude Sonnet 4 ($p = 0.009$) and $\beta_2 = 0.12$ for GPT 4.1 Mini ($p = 0.033$). In other words, as Figure 4b shows, perspective